OLYMPIC INSIGHTS DASHBOARD

Project Documentation | Power BI Data Analytics Case Study

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1. Project Overview

Olympic Insights Using Data Analytics is a Power BI navigational dashboard case study focused on historical Olympic performance. The project turns Olympic records into an interactive analytical experience for examining athlete performance, country-wise medal trends, sport-wise dominance, host-country advantage, gender participation, and athlete demographics.

2. Business Context & Objective

The project report frames the business need as deeper, data-driven visibility into athlete performance, country-wise medal trends, and sport-specific dominance across more than a century of Olympic Games.

Track	Historical achievements of athletes, countries, and sports across Olympic editions.
Evaluate	Medal efficiency, dominance, participation, and host advantage.
Forecast	Use historical patterns as an analytical basis for future planning and athlete-development decisions.

3. Dashboard Architecture

The report describes one Power BI file with six linked pages connected through on-screen navigation buttons.

Page	Dashboard View	Purpose
1	Global Medal Overview	Headline KPIs and top-country medal leaderboard across Olympic Games.
2	Country Insights	Country-level athletes, medals, efficiency, and top sports.
3	Athlete Spotlight	Individual athlete achievements, longevity, and demographics.
4	Sport & Event Analysis	Sports and events driving medal volume and participation.
5	Host Country Analysis	Host-year versus non-host-year performance.
6	Gender & Demographics	Gender balance and athlete age-profile evolution.

4. Data Foundation

The project report describes six linked historical tables: Olympic Athlete Bio, Olympic Athlete Event Results, Olympic Games Medal Tally, Olympic Results, Olympics Country, and Olympics Games. The report states a scope of 1896–2022, 112 sports, 964 distinct events, 231 participating countries/NOCs, and 55 editions held.

The supplied Excel workbook in this project package contains two sheets: **athlete_events** and **country_definitions**. The athlete_events sheet contains 271,116 data records plus its header and 15 columns; country_definitions contains NOC, region, and notes fields. This workbook is retained as the supplied dataset and is documented separately from the broader six-table model described in the report.

5. Key Dashboard KPIs & Findings

Metric	Report value
Total medals awarded	20,654
Countries/NOCs	231

Metric	Report value
Olympic editions held	55
Top country	USA – 3,009 medals
Top sport by medals	Athletics – 4,509
India example	260 medals; Hockey – 216 medals
Athlete medal record	Michael Phelps – 28 medals
Female share	48% in the 2020 Games

6. Tools & Methodology

Data Preparation: Power Query was used to clean and merge six CSV tables, handle null values, parse birth dates, and map historical NOC code changes.

Data Modeling: A star-schema approach was used, with Games and Country as dimension tables and athlete/event and medal data as fact tables.

DAX: Measures were created for Medal Efficiency, Host-Year Flag, YoY Medal Growth, Age Buckets, and Gender Share.

Dashboard Build: Six navigational Power BI pages were designed with bookmark-based buttons and slicers for Year, Country, Sport, and Gender.

7. Sample DAX Measures Documented in the Report

Total Medals: SUM of the total field in the Olympic Games Medal Tally table.

Medal Efficiency: DISTINCTCOUNT of athlete IDs divided by total athletes, multiplied by 100, as documented in the report.

Host-Year Flag: Compares the selected country NOC with the Games host NOC to classify Host or Non-Host.

Athlete Age at Medal: DATEDIFF between athlete birth date and Games start date in years.

YoY Medal Growth: Compares current total medals with the previous Games year.

8. Challenges & Learnings

The project report identifies historical NOC code changes, missing birth dates in older records, team-versus-individual medal counting, and the need to maintain disciplined relationships across a large multi-table model as key challenges.

9. Business Insights & Use Cases

The six dashboard pages provide different decision lenses: national competitiveness benchmarking, country-level sport diversification, athlete scouting and retention, sport investment, host-bid analysis, and gender/age participation monitoring.

10. Conclusion

The project demonstrates an end-to-end Power BI analytics workflow covering data preparation, modeling, DAX, interactive navigation, KPI analysis, historical trends, and business-oriented insights. It converts Olympic history into a structured dashboard for exploring medal leadership, country efficiency, athlete performance, sports, hosting effects, and demographics.

11. Repository Contents

README.md | olympic_Data.xlsx | Dashboard_Screenshot.png | olympic_Project_Documentation.pdf | olympic project Report By Vaishnavi Lahoti-1.pdf